

MOHANA KANNAN RAVIKUMAR

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EDUCATION

University of California San Diego (UCSD)

San Diego, CA

Master of Science in Computer Science and Engineering (with specialization in AI)

Fall 2025 – Present

- Relevant Coursework: Probabilistic Reasoning and Learning, Recommender Systems, Intro to Robotics

Vellore Institute of Technology (VIT)

Vellore, India

Bachelor of Technology in Computer Science and Engineering — CGPA: 9.1/10.0

2019 – 2023

- Relevant Coursework: Machine Learning, Linear Algebra, AI, Image Processing, Operating Systems

SKILLS

Programming Languages: Java, Python, C++, C, SQL, JavaScript

Frameworks: Spring Boot, TensorFlow, PyTorch, Hugging Face, LlamaIndex, LangChain, Angular, ROS

Tools: Git, Jenkins, Splunk, Salesforce, AWS, Maven

Domains: Backend Development, Computer Vision, Deep Learning, NLP, LLM, Reasoning in LLMs, Machine Learning, Inference Optimization, RAG, Full-stack

EXPERIENCE

Citibank

Chennai, India

Software Development Engineer - C10

Jul 2023 – Aug 2025

- **Architected and deployed 14 REST APIs** for the IVR project, enabling seamless integration with retail partners including Best Buy and Home Depot, directly supporting millions of customer account interactions.
- **Engineered an automated monitoring system** that analyzes Splunk logs across 30 microservices, reducing error detection and root cause analysis time by **70%** and enabling proactive issue resolution before customer impact.
- **Built the Salesforce Deployment Request (SFDR) tool** to track release package, reducing build team deployment preparation time by **40%** and eliminating deployment errors through centralized package tracking.
- **Led end-to-end modernization of 8 microservices**, migrating from Java 8 to Java 17 and orchestrating cloud migration from Pivotal Cloud Foundry to OpenShift, ensuring compliance and platform modernization.
- **Owned development of high-impact customer features** including card requests, credit limit increases, and dispute resolution workflows, driving core banking functionality for retail customers.
- **Designed and implemented CI/CD pipelines** using Jenkins and RLM, automating build and deployment processes to reduce manual errors and improve release reliability.
- **Developed scalable microservices architecture** using **Spring Boot** to support Citi's retail banking operations, ensuring high availability and performance.

RESEARCH AND PUBLICATION

Predicting autism in children at an early stage using eye tracking

Dec 2022 – Apr 2023

- Leveraged **deep learning** on **eye-tracking images** to diagnose **autism** in children.
- Utilized **Inception V3** for **feature extraction** and a **Multilayer Perceptron** for **classification**.
- This approach achieved an **accuracy of 87.3%** on classifying children with autism.

Chatbot for product queries — *In collaboration with Siemens*

Feb 2023 – Mar 2023

- Leveraged the **TAPAS transformer model** with a fine-tuned **BERT encoder** for **NLP-based semantic table parsing**, enabling a natural language interface for structured data retrieval and aggregation operations.
- Utilized the **TAPAS tokenizer** for data preprocessing and **table serialization**, successfully converting **pandas DataFrames** into model-compatible formats for **QA (Question-Answering) inference**.

PROJECTS

Fashion Product Description Generator using BLIP2

Dec 2025

- Fine-tuned **BLIP2 vision-language model** on fashion dataset using **LoRA**, achieving enhanced visual description generation through **Visual Question Answering (VQA)** with parameter-efficient fine-tuning.
- Optimized model performance by implementing **LoRA fine-tuning** to reduce trainable parameters while maintaining accuracy, enabling efficient deployment for fashion image captioning and product attribute extraction.

RAG based app to chat with pdfs

Jun 2024

- Developed document QA system using **LlamaIndex** and **Hugging Face**; optimized **Llama2-7B** for domain-specific retrieval tasks.
- Built an end-to-end retrieval pipeline with vector storage and semantic search to enable natural language interaction with PDF documents.

AWARDS AND CERTIFICATIONS

- Special Achiever Award: University Rover Challenge 2021 Aug 2022
- Achiever Award: International Rover Design Challenge (IRDC 2021) Aug 2022
- Runner-up: Intra-Citi Machine Learning Hackathon Feb 2024
- Generative AI Professional Certification: Oracle Cloud Jul 2024